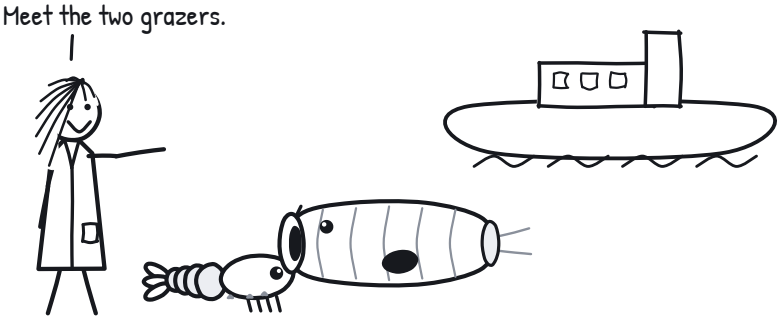
Who Eats What? Krill, Salps and the Southern Ocean Lunch Menu

From "Selective feeding in Southern Ocean key grazers—diet composition of krill and salps" by Nora-Charlotte Pauli, Katja Metfies, Evgeny A. Pakhomov, Stefan Neuhaus, Martin Graeve, Philipp Wenta, Clara M. Flintrop, Thomas H. Badewien, Morten H. Iversen, Bettina Meyer (2021)

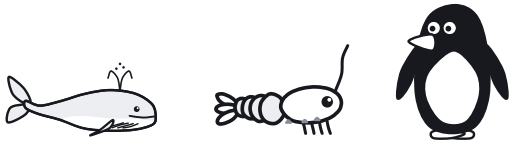
Two grazers

Antarctic krill and the salp Salpa thompsoni are among the dominant grazers in the Southern Ocean.

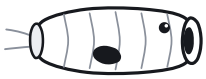
Meet the two grazers.



Krill is a key species in the Southern Ocean ecosystem and an important link between phytoplankton and higher trophic levels such as whales, seals, and penguins.



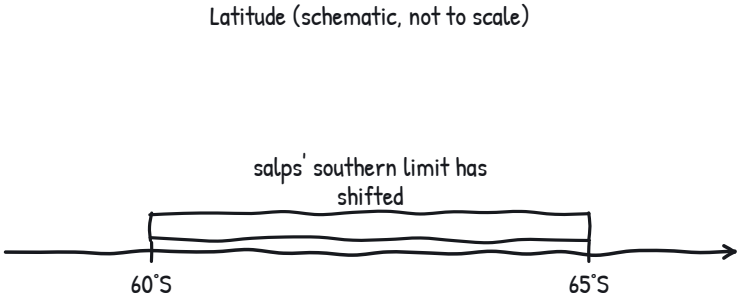
Salps are a less important food source for these apex predators.



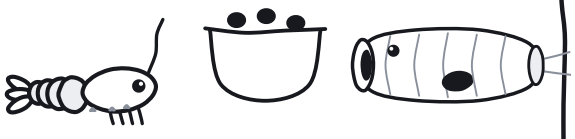
Krill biomass is positively correlated to sea ice and cold temperatures. Opposite to krill, salp abundances are negatively correlated to sea ice cover.



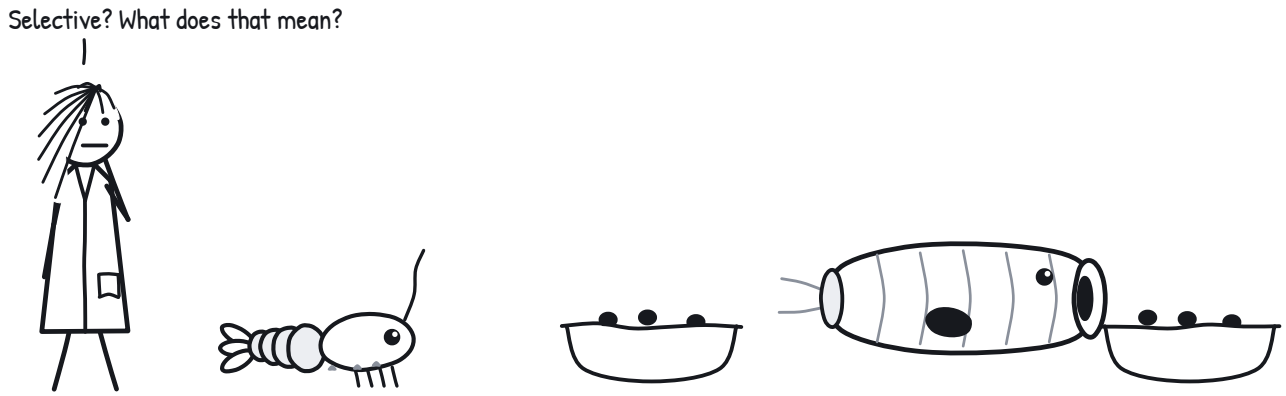
In response to climatic changes, an increase in the abundance of salps was observed and their southern distribution limit has shifted from 60°S to 65°S since 1980.



This has resulted in a greater overlap between the ranges of salps and krill, making them direct competitors for food and habitat in the western Antarctic Peninsula (WAP) region.



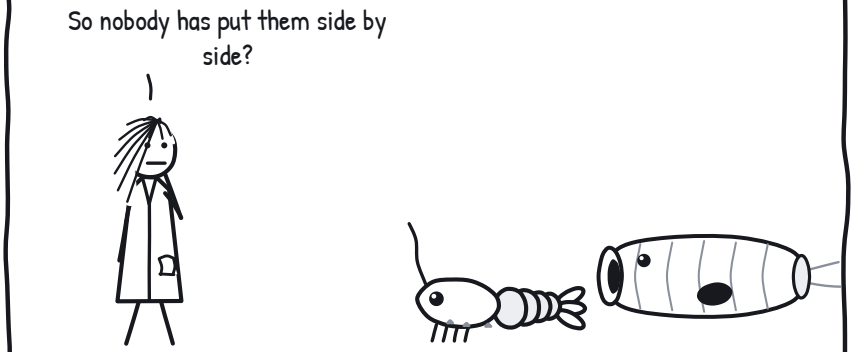
Krill are selective feeders with a diatom-dominated diet. Salps are assumed to be non-selective, indiscriminate feeders.



Selective feeding is the selection of particular prey items while avoiding or rejecting others, and is displayed by most zooplankton.

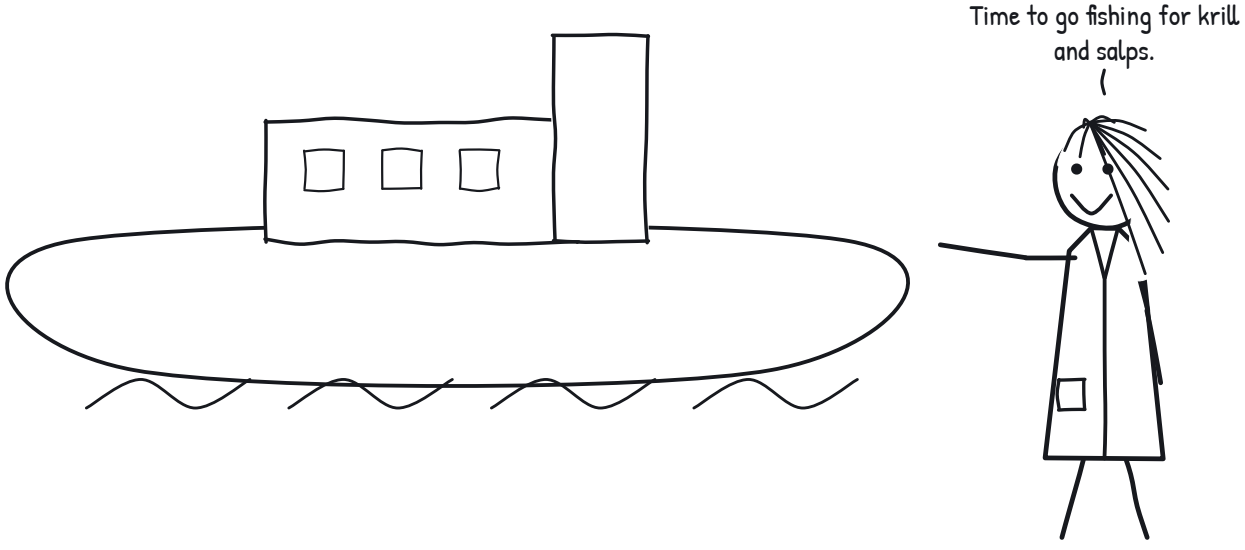


Studies that simultaneously compare the diet composition of krill and salps from the same region are lacking.

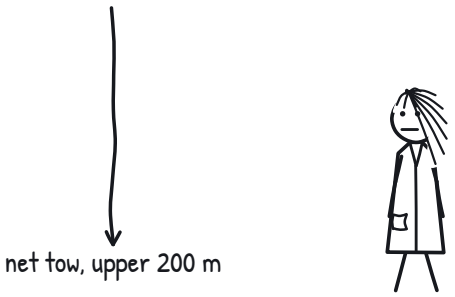


What did they eat?

Samples were collected during the research cruise PS112 with RV Polarstern between March and May 2018 along the Antarctic Peninsula.



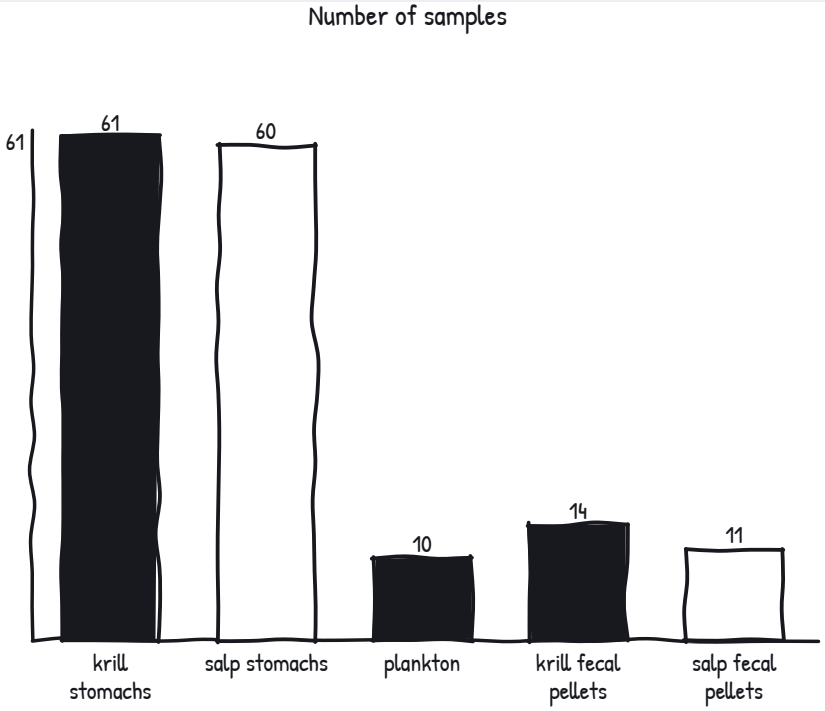
Krill and salps were collected by oblique net tows in the upper 200 m of the water column.



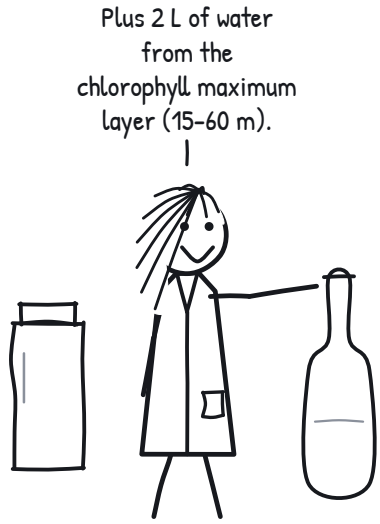
The animals were measured, sexed, staged, frozen in liquid nitrogen, and stored individually at -80 °C within approximately ten minutes.



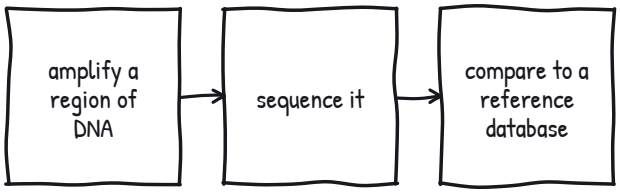
Stomach contents of krill (61 samples) and salps (60 samples) were compared to the ambient plankton community (10 samples). Fecal pellets were also compared (n = 14 for krill, n = 11 for salps).



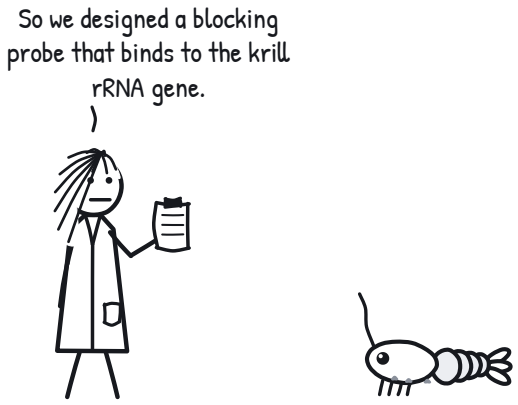
Krill fecal pellets were collected from free drifting sediment traps, and freshly produced salp fecal pellets from incubation experiments.



Metabarcoding is the amplification of a standardized region of the DNA and subsequent sequencing and comparison to a reference database.



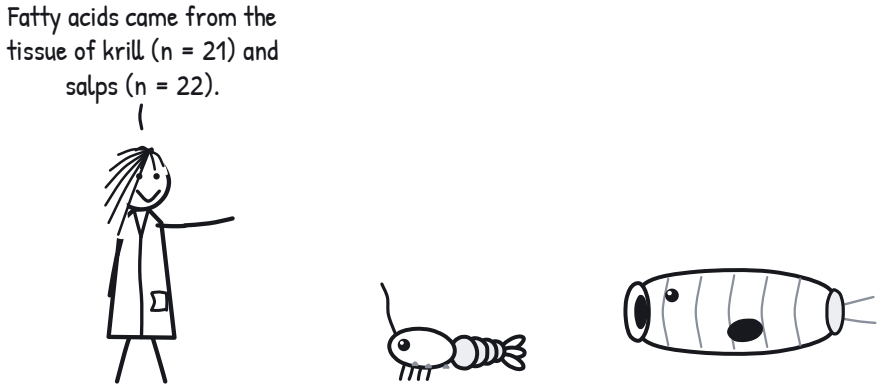
A common methodological issue when sequencing stomach or gut content samples is that predator DNA often swamps prey sequences.



Template controls were randomly included in the library preparation steps, and the authors did not find evidence for contamination.

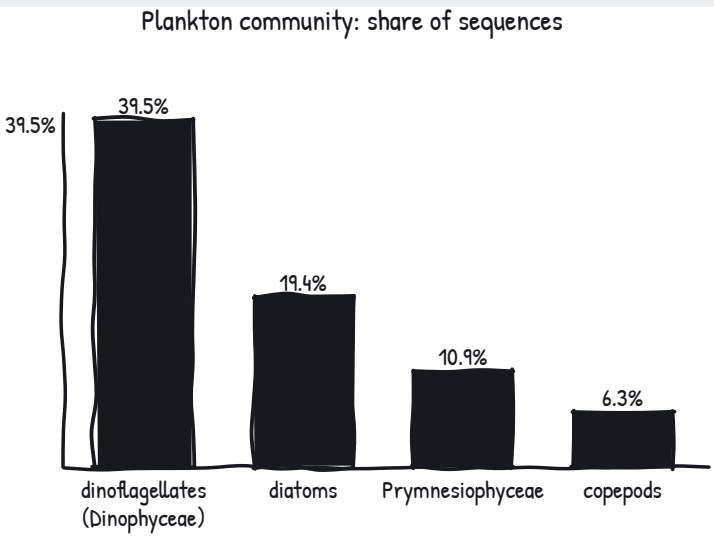


The authors combined 18S rRNA metabarcoding and fatty acid analyses to reveal long- and short-term diets and to identify feeding preferences of both species.

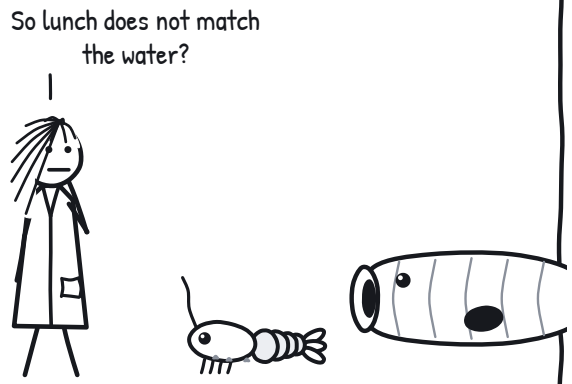


The menu

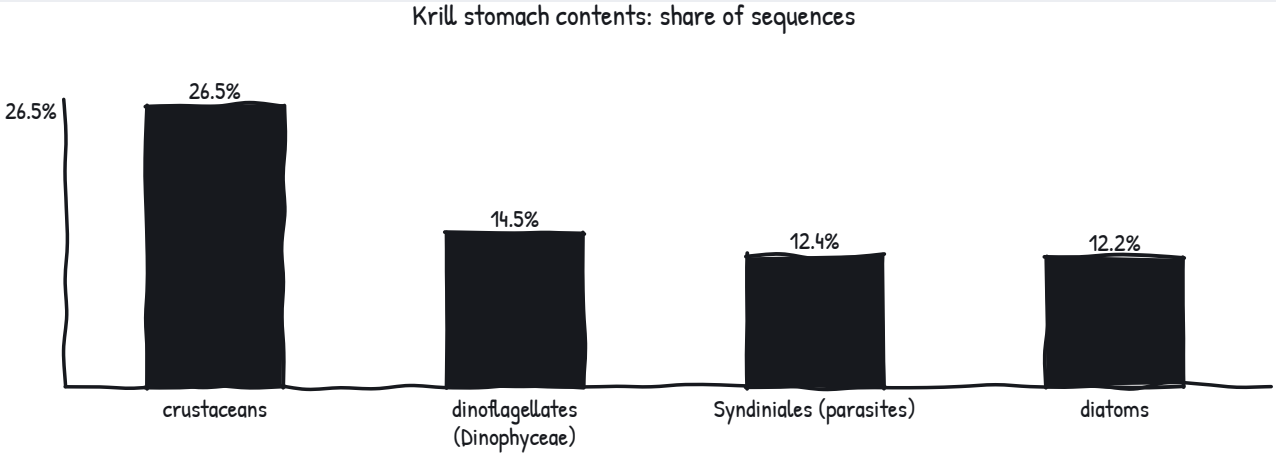
The water column was dominated by dinoflagellates, mainly Gyrodinium (Dinophyceae, 39.5%). Diatoms (19.4%) and Prymnesiophyceae (10.9%) were the most abundant taxa, and copepods accounted for 6.3%.



The sequence libraries from the stomach content of krill and salps did not reflect the dinoflagellate-dominated plankton community nor the high proportion of diatoms observed in the water samples.

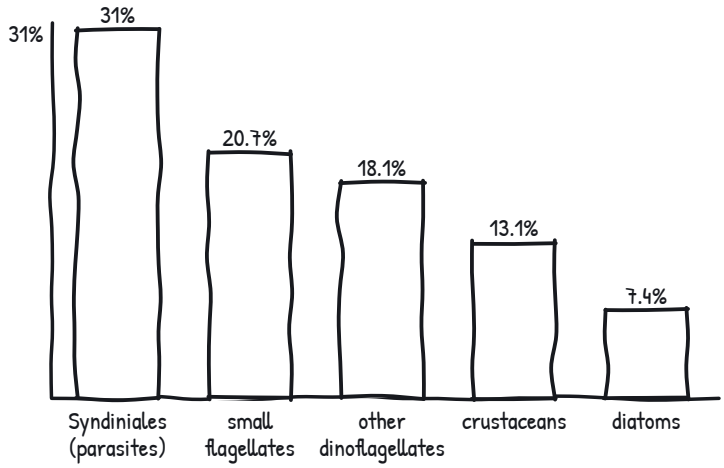


Krill stomachs held a high share of crustacean sequences (26.5%), mainly copepods. Dinoflagellates and diatoms were markedly reduced compared to the plankton (14.5 and 12.2%); the parasitic Syndiniales were enriched (12.4%).

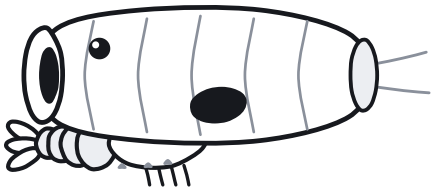


Salp stomachs were dominated by the parasitic Syndiniales (31%), with a high share of small flagellates (20.7%). Other dinoflagellates and diatoms were less abundant than in the plankton (18.1% and 7.4%); crustaceans made up 13.1%.

Salp stomach contents: share of sequences

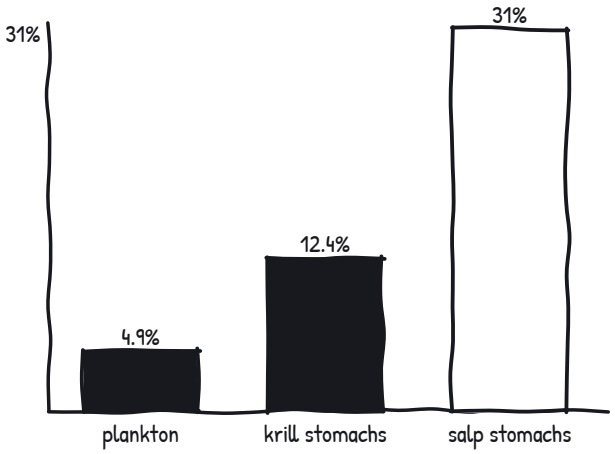


The clusters of stomach content and fecal pellets of both species each largely overlapped.

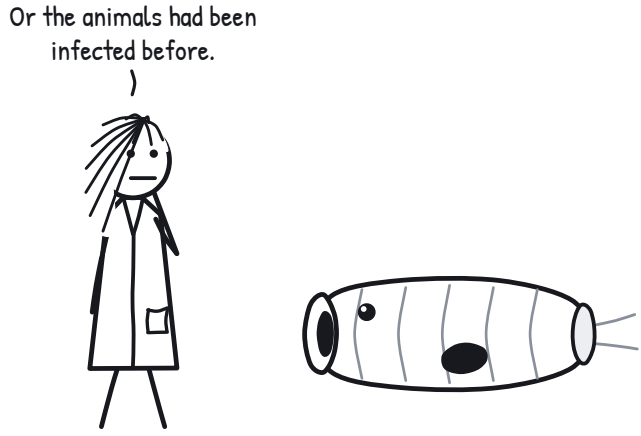


The parasitic dinoflagellate group Syndiniales accounted for almost one third of the salp and 12.4% of the krill diet, while being less abundant in the plankton community (4.9%).

Syndiniales: share of sequences



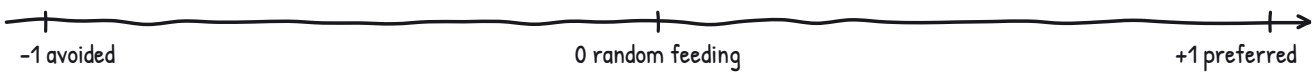
The relative enrichment of Syndiniales sequences in the stomachs of either species versus the plankton community might indicate prior infections of salps or krill.



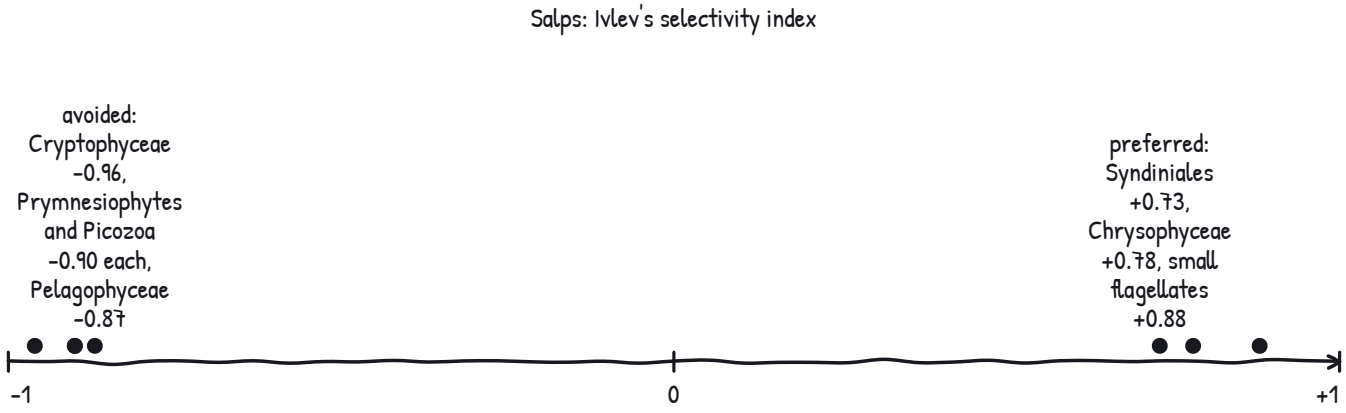
Picky eaters

Ivlev's selectivity index ranges from +1, indicating preference of a prey taxon, to -1, indicating avoidance, while 0 denotes random feeding.

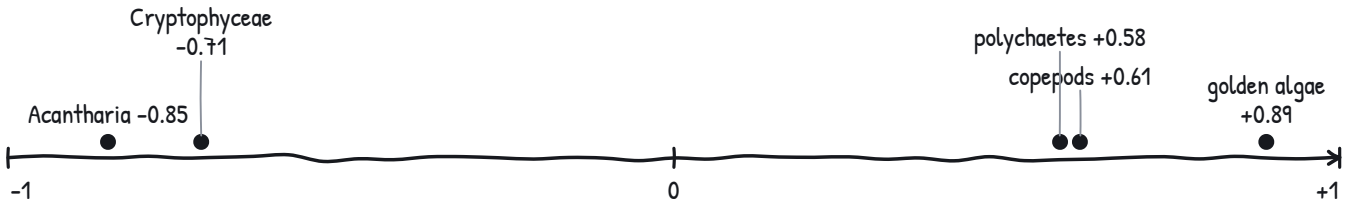
Ivlev's selectivity index



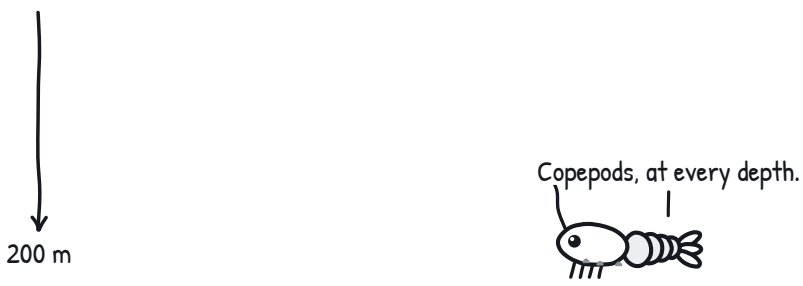
Across all stations, salps showed a strong preference for small flagellates (+0.88), Syndiniales (+0.73), and Chrysophyceae (+0.78), while Prymnesiophytes, Pelagophyceae, Picozoa and Cryptophyceae were avoided.



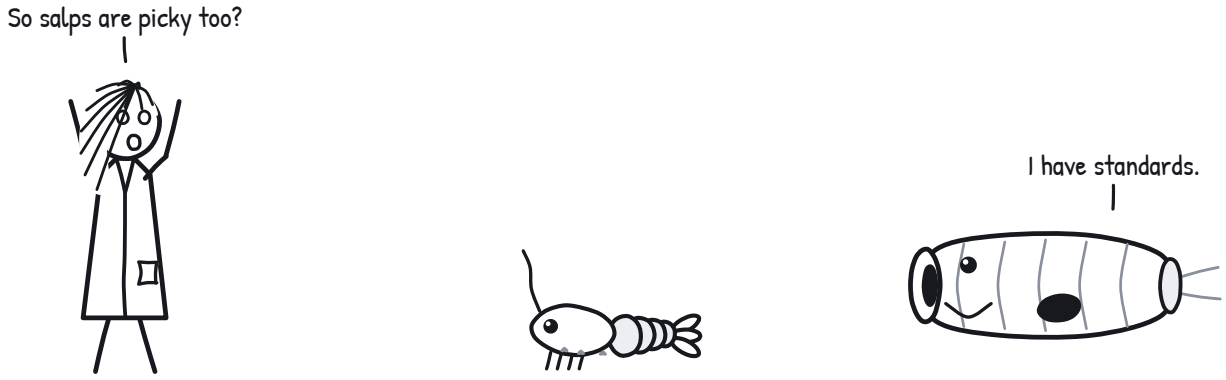
Krill selectively fed on copepods (+0.61), polychaetes (+0.58), and golden algae (Chrysophyceae, +0.89), while avoiding radiolarian protozoans (Acantharia, −0.85) and Cryptophyceae (−0.71).



The feeding selectivity of krill and salps was confirmed across different depths from the surface to 200 m at several stations.

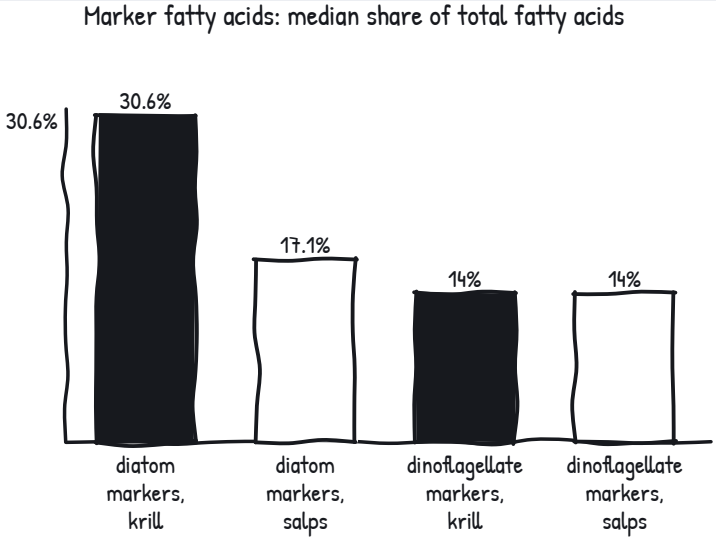


Neither species' diet reflected the composition of the plankton community, suggesting that, in contrast to the accepted paradigm, not only krill but also salps are selective feeders.

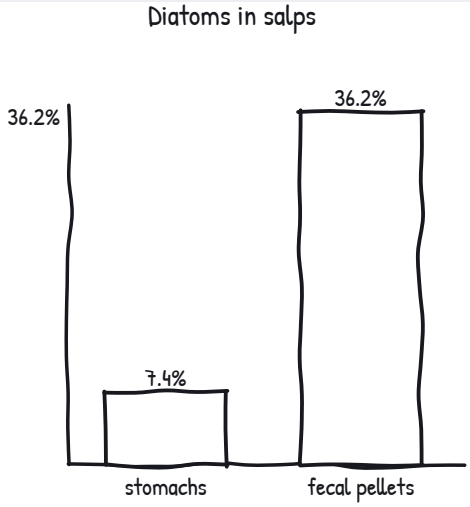


After the meal

Diatom marker fatty acids were higher in krill (median = 30.6% of total fatty acids) than in salps (median = 17.1%). Dinoflagellate marker fatty acids showed no difference between species (median = 14% for each species).



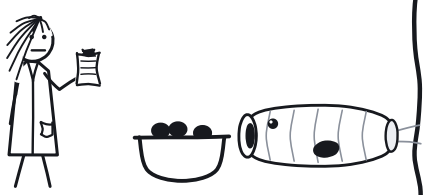
The relative abundance of diatoms was significantly higher in salp fecal pellets (36.2%) compared to their stomachs (7.4%), supporting the notion that salps cannot digest diatoms efficiently.



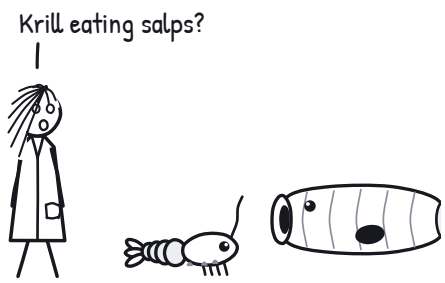
The authors found 17% of diatom fatty acid markers in salp tissue, indicating the assimilation of diatoms.



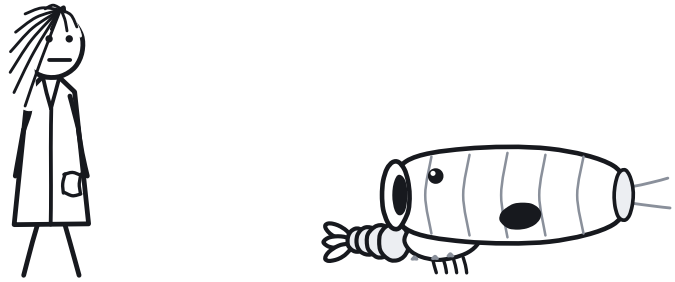
During visual inspections of some salp stomachs on board, the authors observed krill fecal pellets, suggesting that salps ingested those krill fecal pellets.



The diet of krill contained about 6% salp sequences, suggesting that krill were either directly feeding on salps or on their remains.



During on-board video recordings, the authors observed an individual krill holding on to a small chain of salps with its feeding basket and swimming with it for more than one minute before letting go.

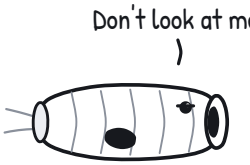


The authors were not able to observe the krill feeding directly on the salps.

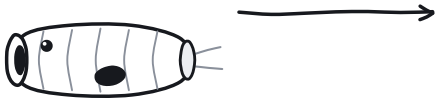


So what?

It remains unclear how salps perform such selective feeding behavior.

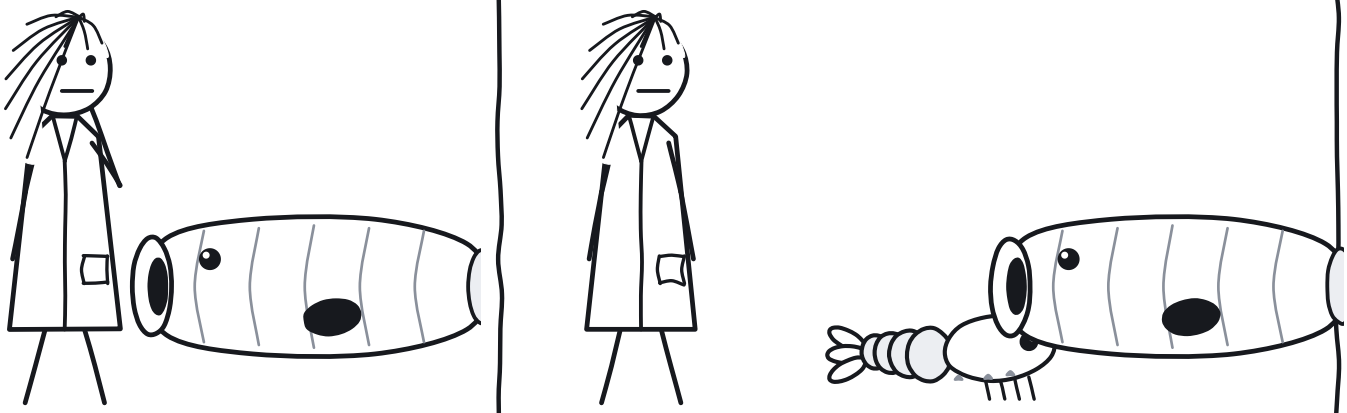


On-board video observations showed that salps are agile swimmers, often performing back-flushes to empty their feeding tract.

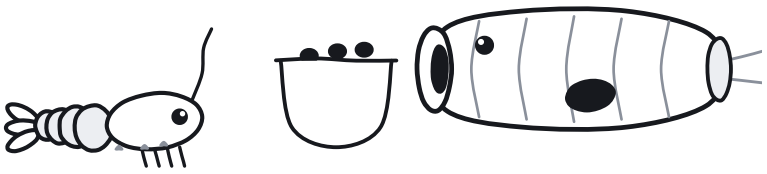


This behavior may allow for selective feeding by rejecting non-desirable food via back-flushing and subsequent swimming to other feeding grounds.

There was no significant difference between krill and salp diet composition, suggesting that, contrary to previous hypotheses, salps might be as selective in their feeding behavior as krill.



Krill and salps had broadly similar diets, potentially enhancing the competition between both species. This could be augmented by salps' ability to rapidly reproduce in favorable conditions, posing further risks to krill populations.



The authors observed selective feeding behavior of both krill and salps and did not find evidence for more efficient feeding on small prey by salps compared to krill. This indicates that the projected shift in plankton size along the WAP might not favor salps as previously suggested.

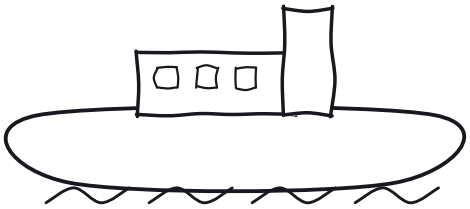


The relative proportion of 18S gene copy numbers is influenced by the relative abundance of the organisms in the sample, as well as by varying gene copy numbers.



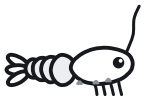
The plankton community was typical for an autumn (March–May) plankton community around the WAP.

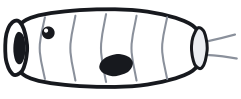




Future studies should further investigate the selective feeding behavior using both field and laboratory tests.







Adapted from "Selective feeding in Southern Ocean key grazers—diet composition of krill and salps" by Nora-Charlotte Pauli, Katja Metfies, Evgeny A. Pakhomov, Stefan Neuhaus, Martin Graeve, Philipp Wenta, Clara M. Flintrop, Thomas H. Badewien, Morten H. Iversen, Bettina Meyer (2021), doi 10.1038/s42003-021-02581-5. Licence: CC BY 4.0. Condensed into a comic with invented characters and scenes using the science-comic-pipeline; the paper's authors did not write or endorse this document.

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